

Changing Perceptions of Development: Pathways to the concept of Sustainability & Sustainable Development

Changing Perceptions of Development

GDP vs GNP:

Gross Domestic Product (GDP):

- Measures the total market value of goods and services produced within a country's borders.
- Includes production by domestic and foreign firms operating in the country.
- Widely used indicator of economic performance.

Formula:

$$\text{GDP} = \text{Consumption} + \text{Investment} + \text{Government Expenditure} + \text{Net Exports}$$

$$\text{GDP} = C + I + G + \text{NX (Net export)}$$

Gross National Product (GNP)

- Measures income earned by a nation's residents regardless of location.
 - Includes earnings from abroad.
 - Excludes income earned domestically by foreign entities.
- $$\text{GNP} = \text{GDP} + (\text{Income Earned by Residents Abroad} - \text{Income Earned by Foreign Residents Domestically})$$

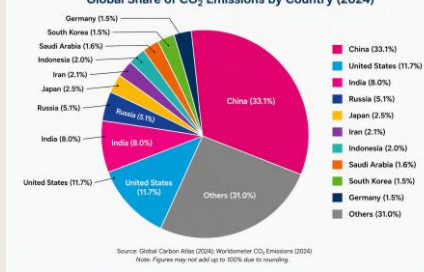
Changing Perceptions of Development

Industrialization, Consumerism and Sustainability:

- Industrialization, consumerism, and capitalism have changed the relationship between society and the environment.
- Globalization has expanded multinational companies and increased consumerism worldwide.
- These trends weaken nature's capacity for recovery and resilience.
- Underdevelopment worsens social inequality.
- Although the rich are the world's main consumers, the poor often suffer the greatest environmental impacts (e.g., climate refugees).
- Experts recognize that **continuous economic growth cannot be sustained indefinitely**.

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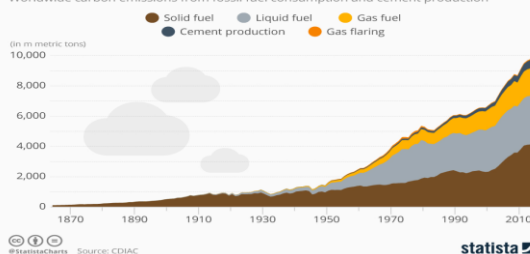
Global Share of CO₂ Emissions by Country (2024)



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The Carbon Age: 150 Years of CO₂ Emissions

Worldwide carbon emissions from fossil fuel consumption and cement production



Changing Perceptions of Development

Green Gross Domestic Product (Green GDP):

Green GDP is an environmentally adjusted measure of economic growth that accounts for the environmental costs of development.

$$\text{Green GDP} = \text{NDP} - \text{Environmental Costs}$$

Environmental costs include:

- Natural resource depletion
- Environmental degradation
- Cost of environmental protection
- CO₂ emissions
- Waste generation (per capita)

Why Green GDP?

- Measures sustainable economic growth
- Supports environmentally responsible policymaking
- Balances economic development with environmental conservation



Changing Perceptions of Development

The Brundtland Commission (1987):



- Established by the United Nations in 1983 and chaired by Gro Harlem Brundtland.
- Identified the growing gap in addressing global environmental problems.
- Emphasized balancing economic development with environmental protection.
- Published Our Common Future (Brundtland Report) in 1987.
- Proposed a long-term strategy for sustainable development.
- Highlighted three key principles:
 - Human Needs
 - Appropriate Technology
 - Intergenerational Equity

Sustainability & Sustainable Development

Sustainability

Sustainability refers to the responsible use of natural resources and human activities in a way that does not compromise the Earth's ecological systems, ensuring a balance between economic growth, environmental protection, and social well-being.

Key Principles

- Protect natural resources and ecosystems.
- Maintain ecological balance.
- Use resources efficiently and responsibly.
- Meet present needs without harming future generations.

Why Can't Economic Growth Continue Indefinitely?

- Natural resources are **finite**.
- Overconsumption leads to **resource depletion**.
- Pollution and climate change threaten ecosystems.
- Environmental degradation reduces the Earth's capacity to support future development.

Sustainability & Sustainable Development

Three Pillars of Sustainability

The three pillars of sustainability are:

1. Economic sustainability
2. Social sustainability, and
3. Environmental sustainability



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1. Economic Sustainability:

- Promotes **long-term economic growth** without depleting natural resources.
- Encourages **responsible resource management** and **fair trade**.
- Supports investment in sustainable infrastructure and innovation.
- Ensures prosperity while minimizing environmental impacts.

2. Social Sustainability

- Promotes **social equity, justice, and human rights**.
- Ensures access to **education, healthcare, and basic services**.
- Supports **inclusive communities**, cultural diversity, and social cohesion.
- Improves the quality of life for present and future generations.

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3. Environmental Sustainability

- Protects and preserves the **natural environment** for present and future generations.
- Promotes the **sustainable use of natural resources**.
- Reduces **pollution** and environmental degradation.
- Encourages the use of **renewable energy** and eco-friendly technologies.
- Conserves **ecosystems and biodiversity**.

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Sustainable Development:

Sustainable Development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

—Brundtland Commission, 1987

Key Features:

- Balances **economic growth, social equity, and environmental protection**.
- Promotes the **sustainable use of natural resources**.
- Encourages **resource conservation, recycling, and reuse**.
- Minimizes **pollution** and environmental degradation.
- Supports **long-term human well-being** while protecting ecosystems.

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Way Forward: Achieving Sustainable Development:

Strategies for a Sustainable Future:

- Build **resilience** through innovative technologies.
- Promote **renewable energy** and clean technologies.
- Create **green jobs** and support sustainable industries.
- Strengthen **research** and evidence-based policymaking.
- Practice the **3Rs**: Reduce, Reuse, and Recycle.
- Conduct **Environmental Impact Assessments (EIA)** before development projects.
- Prevent future ecological damage and restore degraded ecosystems.
- Fulfill our **ethical responsibility** to conserve the Earth for future generations.

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Sustainable Development and ICT:

Negative Impacts of ICT:

- High **energy consumption** throughout the ICT lifecycle.
- Data centers require large amounts of electricity for operation and cooling.
- Increased **carbon emissions** from ICT infrastructure.
- Generation of **non-biodegradable electronic waste (e-waste)**.
- Growing demand for electronic devices increases the use of natural resources.

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Positive Impacts of ICT:

- Enables **smart farming** and environmental monitoring.
- Improves **energy efficiency** and reduces resource consumption.
- Minimizes waste through digital technologies.
- Supports **big data** and GIS for environmental management.
- Helps tackle **climate change** through monitoring and forecasting.
- Promotes the transition to **green technologies**

Thank you all
